



United International University

School of Science and Engineering

Final Examination; Year 2026; Trimester: Spring

Course: PHY 2105; Title: Physics; Sec: All

Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3 are mandatory to answer. Answer any one from question no 4 and 5.

1. (a) One positive and one negative charges with magnitude $+5q$ and $-10q$ are placed on the y axis. The positive charge is on the origin and the negative charge is on the positive y axis at a distance r . Draw electric field lines with appropriate diagrams and propose where (above, or below, or between them) a positive test charge will experience no force. Provide justification of your answer. 2 CO1

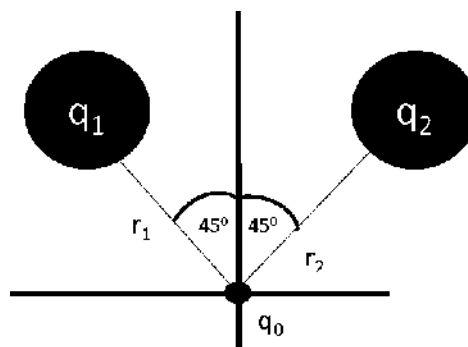
- (b) A dipole is placed in an electric field directed along positive x axis. What is the condition for which the dipole will experience maximum torque and zero torque? Draw appropriate diagrams with the answer. 2 CO1

- (c) The distance between two charges has been increased and now the distance is 2 times compared to the initial distance. What will be the change in force due to the increase in distance? 2 CO1

2. (a) Charges of magnitude 10 nC each are located in vacuum at the corners A, B and C of a right-angle triangle measuring 4 m in height and 2 m base BC. If the charge at A and C are positive and the charge B is negative, what is the magnitude and direction of the total force on the charge at A? 4 CO3

- (b) Three charges lie on the x axis: $q_1 = +5\text{ }\mu\text{C}$, $q_2 = -40\text{ }\mu\text{C}$ and $q_3 = +9\text{ }\mu\text{C}$ at 1 m , 2 m and 5 m respectively from the origin. Calculate (i) magnitude and (ii) direction of net force on a test charge $q_0 = 1\text{ }\mu\text{C}$ at a position $x = 8\text{ m}$. 4 CO3

- (c) The adjacent figure shows two point charges $q_1 = 10\text{ nC}$ and $q_2 = 20\text{ nC}$. What is the magnitude and direction of the net force at the origin on $q_0 = 1\text{ nC}$? Here, $r_1 = r_2 = 2\text{ m}$. 4 CO3

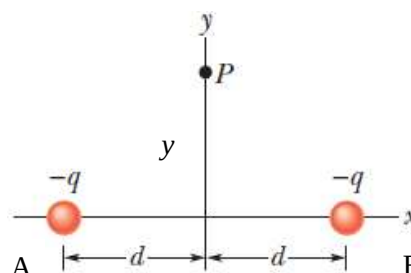


3. (a) An electric dipole consists of charge $+5e$ and $-5e$ separated by 0.4 nm . It is in an electric field of strength $8 \times 10^{-9}\text{ N/C}$. Calculate the magnitude of the potential energy of the dipole when the dipole moment is (i) parallel to and (ii) perpendicular to the electric field. 4 CO3

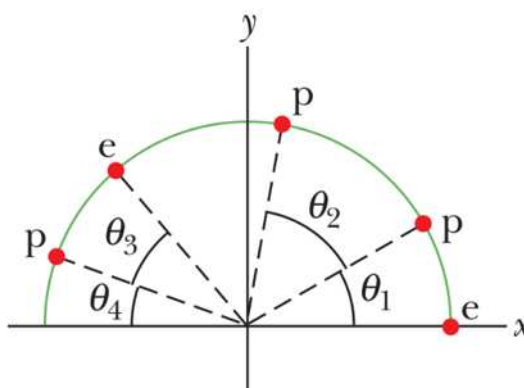
- (b) In the adjacent figure, $q = -8\text{ C}$, $d = 4\text{ m}$ and distance of P from origin is $y = 4\text{ m}$. 6 CO3

- (i) What is the magnitude and direction of the net electric field at point P ?

- (ii) If we replace the $-q$ charge at position A with a charge $Q = +8\text{ C}$, what will be the net electric field (magnitude and direction) at point P now?



(c) Figure shows an uneven arrangement of electrons (e) and protons (p) on a circular arc of radius $r = 4.00$ cm, with angles $\theta_1=30^\circ$, $\theta_2=50^\circ$, $\theta_3=30^\circ$, and $\theta_4=20^\circ$. What is the net electric potential produced at the center of the arc?



4. (a) Show that, the equation of electric field at a point due to a uniformly distributed infinite line charge or ring charge is 4 CO2

$$E = \frac{1}{4\pi\epsilon_0} \frac{qz}{(z^2 + R^2)^{3/2}}$$

where the symbols have their usual meanings.

- (b) Show that, a potential energy U is associated with the orientation of the dipole moment in the field such that, $U = -\vec{P} \cdot \vec{E}$. Calculate the condition for maximum and zero potential energy. Here, $\vec{P}$ is the dipole moment and $\vec{E}$ is the electric field. 4 CO2

5. (a) Show that, the electric potential V due to a continuous charge distribution of length L is 4 CO2

$$V = \frac{\lambda}{4\pi\epsilon_0} \ln\left[\frac{L + (L^2 + d^2)^{1/2}}{d}\right]$$

where the symbols have their usual meanings.

- (b) Show that, at a distance r from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is, 4 CO2

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos\theta}{r^2}.$$

Here, the other symbols have their usual meanings.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.